

AGRI SMART AI

Intelligent Agriculture for Sustainable Farming

28	0.5904	+155.5%	93.6 MB
Supported classes	Model 2 PlantDoc Macro-F1	Relative Macro-F1 gain vs Model 1	Field-adapted checkpoint

Problem

Crop diseases can be difficult to identify reliably from photographs, especially outside controlled laboratory settings. AgriSmart AI turns a leaf image into a practical, farmer-friendly prediction with confidence and precautionary guidance.

Solution

AgriSmart AI uses a field-adapted EfficientNet-B2 computer-vision model to classify crop leaves into **28 disease/healthy classes**. A React + TypeScript interface sends images to a FastAPI service for real-time inference and presents the result in a simple workflow.

Architecture

React + TypeScript Farmer interface	FastAPI Prediction API	EfficientNet-B2 Field-adapted ML
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Data & Method

Training/validation: PlantVillage laboratory images. **External field evaluation:** PlantDoc-style images with natural lighting, clutter and occlusion. The pipeline uses deterministic splits, locked test-set isolation and SHA256-based duplicate checks to prevent leakage.

Key Features

Real-time disease prediction; confidence score; precautionary guidance; responsive farmer-facing UI; English/Hindi/Gujarati interface support; account-specific browser-local profile and scan history.

Model & Evaluation

Model	PlantDoc Macro-F1	Accuracy
Model 1 baseline	0.2311	0.2747
Model 2 field-adapted	0.5904	0.6309

Validation checkpoint

Macro-F1 0.9555 Accuracy 0.9686 Best epoch 17

Why Model 2 matters

The baseline performs very well on clean laboratory data but degrades on realistic field images. Model 2 adapts the classifier using additional field-style training data while keeping the locked evaluation data untouched. The PlantDoc comparison shows a large improvement in field-domain generalization.

Limitations

The classifier is closed-set, so unfamiliar plants or difficult field photographs may still be misclassified. Confidence is a model score, not a guarantee of correctness. The result is informational guidance, not a confirmed agricultural diagnosis. Scan history and profile data are browser-local rather than cloud-synchronized.

Future Scope

Separate prototype modules have also been developed for location-aware weather intelligence, smart irrigation guidance and a farmer assistant. These are **not integrated into the current release** and are reserved for future extension.

Technology

Python, PyTorch, timm, EfficientNet-B2, FastAPI, React, TypeScript, Vite, Tailwind CSS, NumPy, Pandas, Matplotlib.